

Soul meter


**HAN_ UNIVERSITY
OF APPLIED SCIENCES**

Admin manual

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1. Purpose

This manual is for supervisors and developers. It explains how to set up and run the SoulMeter, how to prepare a device to host the server and how to modify the firmware.

2. Basic set up

In order to run the device follow the guide below.

- Set up and start the server on a host device (laptop or computer)
- Plug in and switch the SoulMeter board on.
- Edit questions, points, or other settings if necessary.
- Reset tiles to starting position.
- Do a test run to make sure everything is set up well.

3. Install the server

This guide has been tested for windows and Linux, mac support has not been tested but should work, as it is Linux based.

3.1. Windows install

1. Download and install Node.js LTS from nodejs.org (a)
 - a. Download Link: <https://nodejs.org/en/download>
2. During install, tick “automatically install the necessary tools” so the Bluetooth component can build.
3. Open windows power-shell in the API folder
4. Run command “npm install”.
5. Start the server with command “node index.js”.

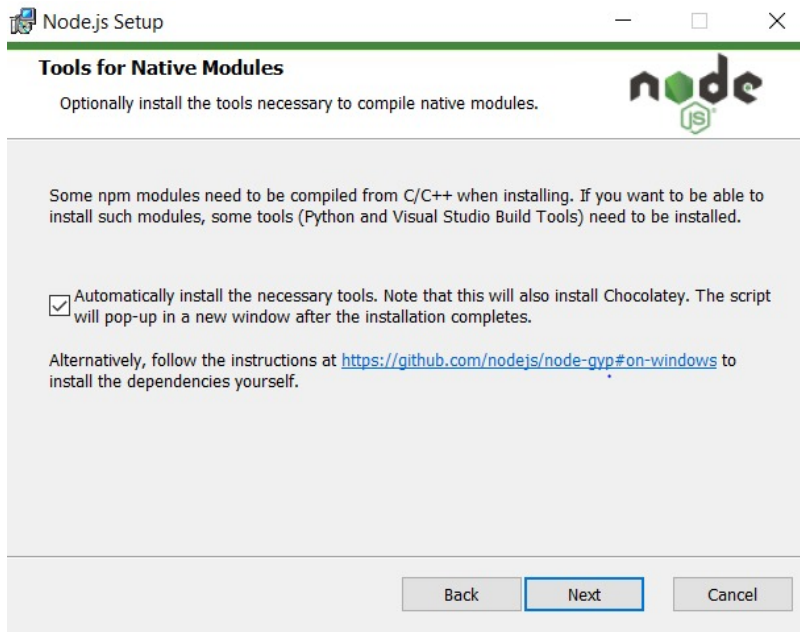


Figure 1: NodeJS, automatically install the necessary tools.

3.2. MacOS install

1. Install Node.js LTS from nodejs.org (a) or with Homebrew (b).
 - a. Download link: <https://nodejs.org/en/download>
 - b. Homebrew command: “brew install node”
2. Open terminal in the API folder.
3. Run command: “npm install”
4. Start the server with command “node index.js
5. Grant Bluetooth permission when macOS prompts it.

3.3. Linux install

1. Install Node.js LTS
 - a. Command: "Sudo apt install nodejs npm"
 - b. apt/dnf/pacman depending on distro.
2. Install build tools
 - a. Command: "sudo apt install build-essential"
3. Open a terminal window in the API folder
4. Run command: "npm install"
5. Start server as root: "sudo node index.js"
 - a. Starting with Sudo prevents issues with bluetooth usage

4. Open the interface

- Game: <http://localhost:3000>
- Admin page (can also be accessed from game): <http://localhost:3000/admin>
- Without the ESP32 (for testing): node index.js —no-ble

5. Edit questions and game settings

- Questions can be edited from the admin page, no code needed. Login using the passcode (1234 by default), change the question, answers, point value, or media link, then save.
- Other settings (port, passcode, starting score, BLE name and timings) are in the api/settings.js file
- Restart the server after editing settings.js

6. Edit ESP32 board code

6.1. One time arduino setup:

1. Install Arduino IDE.
2. Add the Espressif board index: File > preferences > Additional board manager URLs, add the link (a)
 - a. https://raw.githubusercontent.com/espressif/arduino-esp32/gh-pages/package_esp32_index.json
3. Open Tools > Board > Boards manager, search esp32 (by Espressif) and install it
4. Install the FastLED library via Tools > Manage libraries
5. Select the correct port and “ESP32 S3 dev module” as the board.

6.2. Editing and uploading:

1. Open the esp32_board folder in arduino IDE.
2. Adjust values inside config.h (GPIO pins, led colors, LDR threshold value)
3. Press Upload to flash the board

7. Change backlight brightness

- Backlight brightness can be changed using the PWM controller inside the SoulMeter device.
- twist the knob to increase or decrease brightness.



Figure 2: PWM controller module

8. Scores

- Player scores are saved on the host device in `api/scores.csv`
- They can also be downloaded from the running server at <http://localhost:3000/scores/download>